■# PAPER_265: HUDF Dual-Channel Interaction Cascade Buoyancy — Quadratic I(t) Amplification at Cosmic Merger Peak

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__Series:__ Phase 2 Session 72g — §3.x HUDF Clone Fragment Unique Physics Extraction

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■# PAPER_265: HUDF Dual-Channel Interaction Cascade Buoyancy — Quadratic I(t) Amplification at Cosmic Merger Peak

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** HUDFGalaxies.cpp → `HUDFInteractionCascadeTerm` (Session 72g, UQFF 2.0 Upgrade) **Date:** March 2026 **Series:** Phase 2 Session 72g — §3.x HUDF Clone Fragment Unique Physics Extraction

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<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

In the HUDFGalaxies MUGE formulation, the galaxy interaction factor I(t) = I■ · exp(-t/τ_inter) is applied not to one gravitational channel but to **two simultaneously**: the base MUGE term (term1) and the UQFF unification term (term2) both receive the (1 + I(t)) modulation. This creates a structural feature that has not appeared in any prior UQFF module: a **dual-channel interaction cascade** in which both gravity channels amplify coherently during galaxy merger events.

The **uniquely rare discovery** of this paper is that the double application of I(t) produces a **quadratic buoyancy amplification** — the combined effect scales as $(1 + I(t))^2$ rather than the linear (1 + I(t)) of a single-channel system. This cascading enhancement reaches its maximum exactly at t → 0 and z → 3.5, coinciding with the peak observational epoch of the HUDF. The cascade buoyancy excess — purely due to the structural coupling — is ΔI_cascade = $I■^2$ at peak, generating an anomalous buoyancy flux that is second-order in the galaxy interaction strength.

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1. System Parameters

Parameter	Symbol	Value	Units
Peak interaction factor	I■	0.05	—
Interaction timescale	τ_inter	1 Gyr	s
Field mass	M■	10^{12} M_sun	kg
Radius	r	1.23×10^{2}■	m
Average redshift	z_avg	3.5	—
f_TRZ	f_TRZ	0.1	—
Epoch of peak cascade	t	0 (present reference)	s

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2. Dual-Channel I(t) Physics

2.1 Single-Channel Formulation (Baseline)

In a single-channel MUGE, the interaction factor modulates only the base gravity:

```
g_\text{single} = U_{g1} \cdot (1 + H(z) \cdot t) \cdot (1 - B/B_\text{crit}) \cdot (1 +
I(t))
```

The modulation factor is `(1 + I(t)) \to (1 + I_0)` at t = 0. Net relative gain over no-interaction: `+I_0 = +0.05` (5%).

2.2 Dual-Channel Formulation (HUDF Novel)

The HUDF MUGE applies I(t) to BOTH channels:

Channel 1 (base MUGE):

```
\text{term}_1 = U_{g1} \cdot (1 + H(z) \cdot t) \cdot (1 - B/B_\text{crit}) \cdot
\mathbf{(1 + I(t))}
```

Channel 2 (UQFF unification):

```
\text{term}_2 = (U_{g1} + U_{g4}) \cdot (1 + f_\text{TRZ}) \cdot \mathbf{(1 + I(t))}
```

Combined UQFF component at peak (t → 0):

```
g_\text{UQFF,dual}\big|_{t=0} \propto (U_{g1} + U_{g4}) \cdot (1 + f_\text{TRZ}) \cdot (1
+ I_0)^2
```

The `(1 + I_0)^2` factor arises because both channels contribute, and each carries a factor of `(1 + I(t))`. The combined gravity from both terms exceeds the single-channel case by:

```
\Delta_\text{cascade} = I_0^2 \cdot U_{g1} \cdot (1 + f_\text{TRZ})
```

2.3 Cascade Buoyancy Excess

For HUDF parameters at t = 0:

```
I■ = 0.05
U_g1 = G × M■ / r² = 6.674×10■¹¹ × 1.989×10■² / (1.23×10²■)² ≈ 8.77×10■²³ m/s²
(1 + f_TRZ) = 1.1
ΔI_cascade = I■² × U_g1 × (1 + f_TRZ) = 0.0025 × 8.77×10■²³ × 1.1 ≈ 2.41×10■²■ m/s²
```

The cascade excess is ~(I■^{2} / I■) = I■ = 5% of the interaction contribution itself — a second-order but non-negligible buoyancy enhancement at high merger rates.

2.4 Time Evolution: Cascade Peak Alignment with HUDF Epoch

I(t) = I■ · exp(-t/τ_inter):

```
t = 0: I(0) = 0.050 → cascade ΔI = 2.41×10■²■ m/s²
t = 1 Gyr: I(1G) = 0.0184 → cascade ΔI = 3.26×10■²■ m/s² (87% reduction)
t = 2 Gyr: I(2G) = 0.0068 → cascade ΔI = 4.42×10■²■ m/s² (98% reduction)
t = 13.8 Gyr: I(∞) ≈ 0 → cascade ΔI ≈ 0 (local universe quiet)
```

The cascade buoyancy excess is strongly concentrated in the early universe (t < 1 Gyr ≈ z > 3) — precisely the HUDF observational window. This temporal coincidence is **not an artifact**: the f_TRZ > 0 condition ensures UQFF is enhanced in CPT-violating early-universe environments, and, by the cascade mechanism, this enhancement is quadratically sensitive to galaxy merger activity.

3. Cascade Buoyancy Universality Theorem

Theorem (Cascade Buoyancy Universality): In any UQFF system where the interaction factor I(t) is applied to N independent gravitational channels simultaneously, the total buoyancy modulation scales as:

```
\mathcal{B}_N(t) = \prod_{i=1}^N (1 + I(t)) = (1 + I(t))^N
```

The excess buoyancy over single-channel is:

```
\Delta\mathcal{B} = (1 + I)^N - (1 + I) = (1 + I)\left[(1 + I)^{N-1} - 1\right]
```

For N = 2 (HUDF dual-channel), the quadratic interaction enhancement is the minimum non-trivial cascade order. The HUDF is the **first UQFF module proven to be in N = 2 cascade configuration**, establishing that higher-density cosmic environments (more interacting galaxy populations) may activate N > 2 cascade orders.

Corollary: A single ALMA observation of inter-galaxy ^{13}CO molecular bridging between two HUDF merger pairs could constrain I■ to within 10%, directly testing the cascade buoyancy prediction through the expected isotopic enhancement ΔI_cascade.

4. Observational Predictions

- **HST/JWST morphology:** Merger pair fractions at $z \approx 3$–4 (HUDF field) should show anomalously enhanced tidal bridge luminosity proportional to $I■^2$ — cascade-boosted baryonic flow across the gravitational bridge.
- **ALMA Band 3 (3mm CO):** Molecular gas in HUDF $z \approx 3.5$ interacting pairs should show velocity dispersion $\propto (1 + I■)^2$ relative to isolated galaxies of same mass.
- **EHT 345 GHz (future):** Any compact radio core in a HUDF merger would show a DPM resonance fingerprint at the cascade-boosted gravity level.

5. References

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PAPER_265 | UQFF v4.27 | Star-Magic | Session 72g | March 2026